



Privacy



What is privacy

- Privacy is the right to control how your data is collect, used and shared, focusing policy and user right.
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Privacy $\neq$ security

- Privacy is focusing on the management of data
- Security is focusing on preventing measure that can avoid misuse of data
- The critical difference between them is the spread of data is under control and the spread of data is under consent from user



Example for security $\neq$ privacy

- For example:
 - When data is share under consent and protect, privacy and security is preserved
 - When data is share without the user's consent, but it is still under protected, meaning the spread of data is controlled. This case will be consider privacy is compromised and security is preserved
 - When data is share without user's consent and the spread of data is not controlled, both privacy and security is compromised



Technique that we can apply to reinforce privacy

- The aim of these techniques is to remove or hide personal identifiable information, making information that has been shared with consent unusable against users.
- Those techniques include:
 - Data aggregation
 - Homomorphic encryption
 - Differential privacy



Data aggregation

- Data aggregation preserve privacy by grouping the data into a more general, larger interval.
- The aim of data aggregation is hiding individual data point, making the data no longer be personal identifiable.
- One of the most common approach of data aggregation is statistics
 - Statistics does not only provide trend and general overview of how things going
 - From the perspective of cybersecurity, it also provide privacy, because we don't want those sensitive data point to be disclose to public or others who can access in the future.



Homomorphic encryption

- Homomorphic encryption is a special encryption algorithm that allows operation on the data after the data is being encrypted.
- There are various types of homomorphic encryption:
 - Multiplicatively
 - Additively
 - etc
- There is more than these two types of homomorphic encryption, these two are the common ones



Homomorphic encryption

- Additively homomorphic encryption satisfied the property of:
 - $E(m1) + E(m2) = E(m1 + m2)$
- Multiplicatively homomorphic encryption satisfied the property of:
 - $E(m1) \times E(m2) = E(m1 \times m2)$
- It provides the ability to compute on encrypted data, controlling way of information from being shared and used

Differential privacy

- It is perturbation method that protect the privacy of a individual in database, while allowing statistical analysis of data
- It is achieved by adding noise to the output of the dataset.
- After applying differential privacy, if there is a difference in database, the difference is minimised and hardly be notice

